

Paper IV v1.21-r1 — external adversarial revalidation

Run: `run_v1.21_r1` (2026-09-30 23:03 UTC – 2026-10-01). **Target:** `AUDIT_TARGET_v1.21.json` (sha256 574352e8...), editorial candidate `v1.21_editorial_candidate` (manifest abae830f..., ZIP r1 5fc4099a...), unchanged Lean cut `piv-v12-fb459343d234`. **Auditor:** Claude Opus 5.5, running in the same session that produced `run_v1.2_r1`. This revalidation is therefore neither blind nor independent of that run. It is not a cross-family check, and it ran on the same machine and shared dependency cache. **No Lean was run in this cycle.**

Overall verdict: INCONCLUSIVE. v1.21 resolves all MODERATE findings of v1.2 (X-01, X-02, X-03, X-04, X-27), and six of the seven A.2 derivations are now ACCEPTABLE_SUMMARY. One mandatory E2 item is still open: **A2-4 / X-05**. C.3 now defines the schedule and derives the selector chain explicitly. However, the final step is still asserted without proof: the regularity recurrence "iterated h times ... dominates these three quantities after just its first two iterations; absorbing the five terms of (C.1) gives $N_{\text{far}}(\eta_0) \leq T(h+7)$ ". The regularity bound B and its recurrence are not defined anywhere in the manuscript, and A.2 points to a "recurrence in §6.3" that §6.3 does not contain. This affects only the explicit tower values: $T(h+7)$ and $T(h+8)$ in §6.3, and the $c_{\text{far}}/h_{\text{iter}}$ part of (6.28). It affects neither the existence of thresholds nor any theorem. No mathematical error, identity defect, attribution blocker or formal problem was found.

This is an AI review. It is not human peer review and not permission to publish.

1. Gate verdicts

Gate	Verdict	Method	Basis (20_ EVIDENCE/)
E0 Identity	PASS	repeated (start and end identical)	E0/E0_RECORD.md
E1 Claims	PASS (1 MINOR)	repeated on the diff; v1.2 claim map reused	E1/E1_RECORD.md
E2 Mathematics	INCONCLUSIVE	repeated: rederivation of every added passage	E2/E2_RECORD.md
E3 Conformance	PASS	static repetition; v1.2 dynamic results reused (source unchanged)	E3/E3_RECORD.md
E4 Formal reproduction	PASS — reused_verified_external_build	not recompiled; records re-verified	E4/E4_RECORD.md
E5 Falsification	PASS	21 new exact checks repeated; v1.2 T01–T24 reused	E5/E5_RECORD.md
E6 Bilingual	PASS with findings	repeated; 143/143 pages viewed	E6/E6_RECORD.md
E7 Attribution	PASS	repeated; later-version check 2026-09-30	E7/E7_RECORD.md
E8 Author evidence	PASS with observations	repeated (detector re-tested independently)	E8/E8_RECORD.md

2. A.2 rows

Row	v1.2	v1.21
A2-1 regularization	ACCEPTABLE_SUMMARY	ACCEPTABLE_SUMMARY (113/64 sentence added)
A2-2 discard counts	ACCEPTABLE_SUMMARY	ACCEPTABLE_SUMMARY
A2-3 fibres / normalization	REQUIRES_EXPANSION	ACCEPTABLE_SUMMARY. Bilateral cleanup, per-edge load ≤ 1 and retention are derived and were rederived. Observation: the second-moment bad-root bounds are cited, not derived.
A2-4 Table C.1 / tower	REQUIRES_EXPANSION	REQUIRES_EXPANSION (narrowed, MINOR): the regularity recurrence and the final $T(h+7)$ comparison are still missing.
A2-5 normalization	REQUIRES_EXPANSION	ACCEPTABLE_SUMMARY (counts derived)
A2-6 (E.2b)/(E.2d)	REQUIRES_EXPANSION	ACCEPTABLE_SUMMARY (joint elimination and the n/9 error derived)
A2-7 Bose/Skolem	ACCEPTABLE_SUMMARY	ACCEPTABLE_SUMMARY

3. Findings matrix (full list in FINDINGS.csv)

- **Resolved:** X-01, X-02, X-03, X-04, X-06, X-07, X-08, X-09, X-10, X-11, X-13, X-14, X-16, X-20, X-21, X-23, X-25, X-27.
- **Partial:** X-05 (MINOR, open step above). X-12 (MINOR: `E34.lemma3` and `E34.transfer` are still plain text). X-15 (MINOR: "matching(s)" $\times 12$ and "packing" $\times 1$ in ES prose; "baseline" in ES Figure 3).

- **Mitigated:** X-28. The external detector was verified against the real Lean 4.28 warning and against variants, and finds 0 hits in all 1266 logs. The frozen runners stay unpatched by design.
- **Kept, as declared:** X-17, X-18, X-19, X-22, X-24, X-26, X-29, AUD-P1, AUD-C1, AUD-C2.
- **New:**
 - NEW-01 (MINOR): descriptive words set as code (`Regularization` in EN §5.1 and A.2; `Nibble` in ES Lemma 3.2), with EN/ES heading inconsistency.
 - NEW-02 (MINOR): the ES reference [6] title is corrupted to «Integer and fractional `empaquetamientos` in dense graphs».
 - Observations R-01, R-02, E8-01, E8-02, E8-03.

4. What was repeated vs reused

- **Repeated in this cycle:**
 - E0 identities (twice).
 - The auditor's own v1.2→v1.21 diff and rederivation of all added mathematics.
 - 21 exact numeric checks with negative controls.
 - Identifier resolution against the frozen source.
 - The X-27 text check against `AuditorASProbe.log`.
 - Later-version literature checks.
 - Independent detector tests and a rescan of all 1266 logs.
 - Re-verification of the E4 records without trusting the overwritten `SUMMARY.json`.
 - A full page-by-page visual inspection.
- **Reused, not repeated:** the E4 build and its cone/axiom/olean evidence, the 24 E5 tests of v1.2, and the v1.2 E3 dynamic checks. All are justified by the unchanged sources and by E0.

5. Limitations

- Same session and same model as the audited predecessor run. Same family as the manuscript's assistant.
- Same machine and shared pinned Mathlib oleans. No independent kernel replay.
- The E6 automated parity checks were run by a same-family subagent that was stopped twice by rate limits. The lead auditor did all the page viewing, on 90 dpi 2-up renders.
- Literature coverage as in `run_v1.2_r1`. No complete novelty search.

6. Recommendation (recommendation only)

1. To close A2-4/X-05, add to C.3:

- the regularity bound $B(\delta/8, k_0)$ used and its recurrence;
- the two-iteration domination of $G_1 \dots G_3$;
- the absorption of the five (C.1) terms into $T(h+7)$.

Then fix the A.2 sentence that points to "the recurrence in §6.3". Alternatively, state $T(h+7)$ and $T(h+8)$ explicitly as formally verified values whose comparison is not reproduced in the text.

2. Correct the ES reference [6] title (NEW-02), the residual X-12 and X-15 items, and NEW-01.

3. Any corrected text needs a new identity. A short revalidation of E0/E2 (C.3 only)/E6 would then suffice. No Lean rebuild is needed while the cut stays unchanged.